200 APTITUDE QUESTIONS WITH ANSWERS & EXPLANATIONS

Basic + Advanced | Job Interview Preparation

SECTION 1 — BASIC APTITUDE

1. What is 25% of 240?

Answer: 60

Explanation:
25% = 25/100 = 1/4
240 × 1/4 = 60

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2. A number increased by 20% becomes 360. What was the original number?

Answer: 300

Explanation:
After a 20% increase, the number becomes 120% of the original.
Original × 120/100 = 360
Original = 360 × 100/120 = 300

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3. The ratio of boys to girls is 3:5. If there are 24 boys, how many girls are there?

Answer: 40

Explanation:
3 parts = 24
1 part = 24 ÷ 3 = 8
Girls = 5 × 8 = 40

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4. Find the average of 18, 24, 30, 36 and 42.

Answer: 30

Explanation:
Average = Sum ÷ Number of values
= (18 + 24 + 30 + 36 + 42) ÷ 5
= 150 ÷ 5 = 30

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5. A shirt costs ₹800 and is sold at a 15% discount. Find the selling price.

Answer: ₹680

Explanation:
Discount = 15% of ₹800 = ₹120
Selling Price = ₹800 − ₹120 = ₹680

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6. An item bought for ₹500 is sold for ₹575. Find the profit percentage.

Answer: 15%

Explanation:
Profit = 575 − 500 = ₹75
Profit% = (75/500) × 100 = 15%

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7. Find the simple interest on ₹6,000 at 8% per annum for 2 years.

Answer: ₹960

Explanation:
SI = (P × R × T)/100
= (6000 × 8 × 2)/100
= ₹960

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8. A train travels 180 km in 3 hours. What is its average speed?

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Answer: 60 km/h

Explanation:
Speed = Distance ÷ Time
= 180 ÷ 3 = 60 km/h

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9. If 8 workers finish a job in 15 days, how many worker-days are required?

Answer: 120 worker-days

Explanation:
Worker-days = Workers × Days
= 8 × 15 = 120 worker-days

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10. What is 3/5 of 250?

Answer: 150

Explanation:
250 × 3/5 = 50 × 3 = 150

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11. Find the HCF of 36 and 48.

Answer: 12

Explanation:
Factors of 36 include 1, 2, 3, 4, 6, 9, 12...
Factors of 48 include 1, 2, 3, 4, 6, 8, 12...
Highest common factor = 12

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12. Find the LCM of 12 and 18.

Answer: 36

Explanation:
Multiples of 12: 12, 24, 36...

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Multiples of 18: 18, 36...
Therefore, LCM = 36

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13. What is the remainder when 157 is divided by 12?

Answer: 1

Explanation:
12 × 13 = 156
157 − 156 = 1

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14. If x + 17 = 42, find x.

Answer: 25

Explanation:
x = 42 − 17 = 25

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15. Solve: 3x = 81.

Answer: 27

Explanation:
x = 81 ÷ 3 = 27

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16. What is 0.75 expressed as a fraction in simplest form?

Answer: 3/4

Explanation:
0.75 = 75/100
Divide by 25: 75/100 = 3/4

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17. Convert 7/8 into a percentage.

Answer: 87.5%

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Explanation:
(7/8) × 100 = 87.5%

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18. A 20% loss on an item gives a selling price of ₹960. Find the cost price.

Answer: ₹1,200

Explanation:
After 20% loss, SP = 80% of CP.
80% of CP = 960
CP = 960 × 100/80 = ₹1,200

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19. If 5 pens cost ₹75, what is the cost of 12 pens?

Answer: ₹180

Explanation:
Cost of 1 pen = 75 ÷ 5 = ₹15
Cost of 12 = 15 × 12 = ₹180

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20. A car uses 8 litres of fuel for 120 km. How much fuel is needed for 300 km?

Answer: 20 litres

Explanation:
Fuel per km = 8/120
For 300 km = 8 × 300/120 = 20 litres

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21. Find the next number: 2, 5, 8, 11, __.

Answer: 14

Explanation:
Each number increases by 3.
11 + 3 = 14

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22. Find the next number: 3, 6, 12, 24, __.

Answer: 48

Explanation:
Each number is multiplied by 2.
24 × 2 = 48

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23. If 40% of a number is 72, find the number.

Answer: 180

Explanation:
40% of x = 72
x = 72 × 100/40 = 180

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24. A person's salary is ₹30,000. It increases by 10%. What is the new salary?

Answer: ₹33,000

Explanation:
Increase = 10% of ₹30,000 = ₹3,000
New salary = ₹30,000 + ₹3,000 = ₹33,000

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25. Find the average of the first five positive even numbers.

Answer: 6

Explanation:
Numbers = 2, 4, 6, 8, 10
Average = 30 ÷ 5 = 6

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26. A can do a job in 12 days. What fraction of the job does he complete in one day?

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Answer: 1/12

Explanation:
If the complete work takes 12 days, work done in one day = 1/12

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27. A clock shows 3:00. What is the angle between the hands?

Answer: 90°

Explanation:
At 3:00, the minute hand is at 12 and hour hand is at 3.
The angle between them is 90°.

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28. How many degrees are in a straight angle?

Answer: 180°

Explanation:
A straight line forms an angle of 180°.

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29. What is the perimeter of a square with side 9 cm?

Answer: 36 cm

Explanation:
Perimeter = 4 × side
= 4 × 9 = 36 cm

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30. Find the area of a rectangle of length 12 cm and breadth 7 cm.

Answer: 84 cm²

Explanation:
Area = Length × Breadth
= 12 × 7 = 84 cm²

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31. Find the area of a triangle with base 10 cm and height 8 cm.

Answer: 40 cm²

Explanation:
Area = 1/2 × Base × Height
= 1/2 × 10 × 8 = 40 cm²

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32. A box has 4 red and 6 blue balls. What is the probability of drawing a red ball?

Answer: 2/5

Explanation:
Total balls = 10
Red balls = 4
Probability = 4/10 = 2/5

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33. What is 15^2?

Answer: 225

Explanation:
15 × 15 = 225

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34. Find $\sqrt{144}$.

Answer: 12

Explanation:
12 × 12 = 144
Therefore, $\sqrt{144} = 12$

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35. What is 2^5?

Answer: 32

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Explanation:
$2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$

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36. If $4x + 3 = 27$, find x.

Answer: 6

Explanation:
$4x = 27 - 3 = 24$
$x = 24 \div 4 = 6$

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37. The sum of two numbers is 50 and their difference is 14. Find the larger number.

Answer: 32

Explanation:
Larger number = (Sum + Difference)/2
= (50 + 14)/2 = 32

---

38. A number is divisible by both 6 and 8. What is the least possible positive number?

Answer: 24

Explanation:
LCM of 6 and 8 = 24

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39. Find the median of 4, 7, 9, 12, 15.

Answer: 9

Explanation:
There are 5 values arranged in order.
The middle value is 9.

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40. Find the mode of 2, 3, 3, 5, 7, 3, 8.

Answer: 3

Explanation:
3 appears three times, more than any other number.
Mode = 3

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41. A shopkeeper marks an item at ₹1,000 and gives 10% discount. Find the discount amount.

Answer: ₹100

Explanation:
Discount = 10% × 1000 = ₹100

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42. If 12% GST is charged on ₹2,500, what is the GST amount?

Answer: ₹300

Explanation:
GST = 12% × 2500
= ₹300

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43. Convert 2.5 hours into minutes.

Answer: 150 minutes

Explanation:
1 hour = 60 minutes
2.5 × 60 = 150 minutes

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44. A bus covers 240 km at 48 km/h. How long does it take?

Answer: 5 hours

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Explanation:
Time = Distance ÷ Speed
= 240 ÷ 48 = 5 hours

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45. If 6 machines make 900 units in a day, how many units will 10 machines make at the same rate?

Answer: 1,500

Explanation:
One machine = 900 ÷ 6 = 150 units
10 machines = 150 × 10 = 1,500 units

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46. A pipe fills a tank in 10 hours. What part of the tank does it fill in 4 hours?

Answer: 2/5

Explanation:
One hour = 1/10 tank
4 hours = 4/10 = 2/5

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47. If 7 notebooks cost ₹245, find the cost of one notebook.

Answer: ₹35

Explanation:
245 ÷ 7 = ₹35

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48. What is the sum of the first 10 natural numbers?

Answer: 55

Explanation:
Formula = n(n+1)/2
= 10 × 11/2 = 55

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49. What is the probability of getting an even number on a fair die?

Answer: 1/2

Explanation:
Even outcomes = 2, 4, 6 → 3 outcomes
Total outcomes = 6
Probability = 3/6 = 1/2

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50. If a number is increased from 80 to 100, what is the percentage increase?

Answer: 25%

Explanation:
Increase = 100 − 80 = 20
Percentage increase = 20/80 × 100 = 25%

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51. A man spends 70% of his income. If he saves ₹9,000, what is his income?

Answer: ₹30,000

Explanation:
Savings = 30% of income.
30% of income = 9,000
Income = 9,000 × 100/30 = ₹30,000

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52. A mixture contains milk and water in the ratio 4:1. If total mixture is 25 L, how much water is there?

Answer: 5 L

Explanation:
Total parts = 4 + 1 = 5
One part = 25/5 = 5 L
Water = 5 L

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53. A father is 40 years old and his son is 10. What is their age ratio?

Answer: 4:1

Explanation:
40:10 = divide both by 10 = 4:1

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54. If 3 dozen eggs cost ₹216, what is the cost of 5 dozen?

Answer: ₹360

Explanation:
3 dozen = ₹216
1 dozen = ₹72
5 dozen = 72 × 5 = ₹360

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55. A number divided by 7 gives quotient 9 and remainder 4. Find the number.

Answer: 67

Explanation:
Number = Divisor × Quotient + Remainder
= 7 × 9 + 4 = 67

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56. Find 18% of 450.

Answer: 81

Explanation:
450 × 18/100 = 81

---

57. A 10% discount followed by another 10% discount gives what total effective discount?

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Answer: 19%

Explanation:
After first discount, price = 90%.
Second discount leaves 90% of 90% = 81%.
Effective discount = 100 − 81 = 19%

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58. If CP is ₹1,200 and profit is 25%, find SP.

Answer: ₹1,500

Explanation:
Profit = 25% of 1200 = ₹300
SP = 1200 + 300 = ₹1,500

---

59. Find the compound interest on ₹1,000 at 10% per annum for 2 years.

Answer: ₹210

Explanation:
Amount = $1000 \times (1.10)^2$ = ₹1,210
CI = 1210 − 1000 = ₹210

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60. A train 150 m long passes a pole in 10 seconds. Find its speed.

Answer: 15 m/s

Explanation:
Speed = Distance ÷ Time
= 150 ÷ 10 = 15 m/s

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61. Convert 15 m/s to km/h.

Answer: 54 km/h

Explanation:
To convert m/s to km/h, multiply by 18/5.
15 × 18/5 = 54 km/h

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62. A can finish work in 20 days and B in 30 days. How much work do they do together in one day?

Answer: 1/12

Explanation:
A's rate = 1/20
B's rate = 1/30
Together = 1/20 + 1/30
= 3/60 + 2/60 = 1/12

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63. If x:y = 4:7 and y = 35, find x.

Answer: 20

Explanation:
7 parts = 35
1 part = 5
x = 4 × 5 = 20

---

64. Find the sum of angles in a triangle.

Answer: 180°

Explanation:
The three interior angles of every triangle add up to 180°.

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65. A circle has radius 7 cm. Using π = 22/7, find its circumference.

Answer: 44 cm

Explanation:
Circumference = 2πr
= 2 × 22/7 × 7 = 44 cm

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66. Find the volume of a cube with side 5 cm.

Answer: 125 cm^3

Explanation:
Volume = $side^3$
= 5^3 = 125 cm^3

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67. What is the smallest prime number?

Answer: 2

Explanation:
2 is the smallest positive integer having exactly two factors: 1 and 2.

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68. How many prime numbers are there between 10 and 20?

Answer: 4

Explanation:
Prime numbers are 11, 13, 17 and 19.
Total = 4

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69. Find the greatest 3-digit number divisible by 9.

Answer: 999

Explanation:
999 is a 3-digit number and 9 + 9 + 9 = 27, which is divisible by 9.
Therefore = 999

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70. What is the square of 35?

Answer: 1,225

Explanation:
35 × 35 = 1,225

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71. If 5x − 10 = 40, find x.

Answer: 10

Explanation:
5x = 50
x = 10

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72. Find the average of 10 numbers whose sum is 450.

Answer: 45

Explanation:
Average = 450 ÷ 10 = 45

---

73. A student scores 72 out of 90. Find the percentage.

Answer: 80%

Explanation:
Percentage = 72/90 × 100 = 80%

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74. If 2/3 of a number is 50, find the number.

Answer: 75

Explanation:
2x/3 = 50
x = 50 × 3/2 = 75

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75. A product costs ₹2,000. After a 15% increase, what is its price?

Answer: ₹2,300

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Explanation:
Increase = 15% of 2000 = ₹300
New price = ₹2,300

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76. If 4 men can do a job in 18 days, how many days will 6 men take?

Answer: 12 days

Explanation:
Total work = 4 × 18 = 72 man-days
Days for 6 men = 72 ÷ 6 = 12 days

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77. A boat travels 30 km downstream in 2 hours. Find its downstream speed.

Answer: 15 km/h

Explanation:
Speed = 30 ÷ 2 = 15 km/h

---

78. If the average of 6 numbers is 14, find their total.

Answer: 84

Explanation:
Total = Average × Number of values
= 14 × 6 = 84

---

79. Find the next number: 1, 4, 9, 16, __.

Answer: 25

Explanation:
These are squares: 1^2, 2^2, 3^2, 4^2.
Next = 5^2 = 25

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80. Find the next number: 81, 27, 9, 3, __.

Answer: 1

Explanation:
Each number is divided by 3.
3 ÷ 3 = 1

---

81. What is the remainder when 999 is divided by 10?

Answer: 9

Explanation:
999 = 99 × 10 + 9
Remainder = 9

---

82. If 30% of x = 45, find x.

Answer: 150

Explanation:
x = 45 × 100/30 = 150

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83. A shopkeeper buys an item for ₹800 and wants 20% profit after giving 10% discount. What marked price is needed?

Answer: ₹1,066.67

Explanation:
Required SP = 800 × 120/100 = ₹960.
After 10% discount, SP = 90% of MP.
MP = 960/0.90 = ₹1,066.67

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84. Find the simple interest on ₹10,000 at 5% for 3 years.

Answer: ₹1,500

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Explanation:
SI = 10,000 × 5 × 3/100 = ₹1,500

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85. A car travels 150 km in 2.5 hours. Find its speed.

Answer: 60 km/h

Explanation:
Speed = 150 ÷ 2.5 = 60 km/h

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86. If 5 workers build a wall in 24 days, how many days will 8 workers take?

Answer: 15 days

Explanation:
Work = 5 × 24 = 120 worker-days
Days = 120 ÷ 8 = 15 days

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87. The ratio 45 minutes to 2 hours is what?

Answer: 3:8

Explanation:
2 hours = 120 minutes
45:120 = divide by 15 = 3:8

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88. A bag contains 5 white and 15 black balls. Probability of white is?

Answer: 1/4

Explanation:
Total = 20
White = 5
Probability = 5/20 = 1/4

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89. Find the median of 5, 8, 12, 15, 20, 22, 30.

Answer: 15

Explanation:
There are 7 numbers.
The 4th number is the median = 15

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90. What is 40% of 2.5 kg?

Answer: 1 kg

Explanation:
2.5 × 40/100 = 1 kg

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91. A number is increased by 25% and becomes 250. Find original number.

Answer: 200

Explanation:
125% of original = 250
Original = 250 × 100/125 = 200

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92. If 3 pencils and 2 pens cost ₹70, and each pencil costs ₹10, find the cost of one pen.

Answer: ₹20

Explanation:
3 pencils = ₹30
2 pens = 70 − 30 = ₹40
1 pen = ₹20

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93. A person walks at 5 km/h for 36 minutes. How far does he walk?

Answer: 3 km

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Explanation:
36 minutes = 36/60 = 0.6 hours
Distance = 5 × 0.6 = 3 km

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94. Find the area of a circle of radius 7 cm using π = 22/7.

Answer: 154 cm²

Explanation:
Area = πr²
= 22/7 × 7 × 7 = 154 cm²

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95. If the perimeter of a square is 64 cm, find its area.

Answer: 256 cm²

Explanation:
Side = 64/4 = 16 cm
Area = 16² = 256 cm²

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96. A class has 18 boys and 12 girls. What percentage are girls?

Answer: 40%

Explanation:
Total = 30
Girls = 12
Percentage = 12/30 × 100 = 40%

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97. Find the product of the first three prime numbers.

Answer: 30

Explanation:
First three primes = 2, 3, 5
2 × 3 × 5 = 30

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98. What is the value of 1001 × 999?

Answer: 999999

Explanation:
Use $(a+b)(a-b) = a^2-b^2$.
$1001 \times 999 = (1000+1)(1000-1)$
$= 1{,}000{,}000 - 1 = 999{,}999$

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99. If a:b = 2:3 and b:c = 4:5, find a:c.

Answer: 8:15

Explanation:
Make b common.
2:3 → 8:12
4:5 → 12:15
Therefore a:c = 8:15

SECTION 2 — ADVANCED APTITUDE

100. A sum becomes ₹13,310 in 2 years at 15% compound interest per annum. Find the principal.

Answer: ₹10,000

Explanation:
$A = P(1 + R/100)^2$
$13{,}310 = P(1.15)^2$
$13{,}310 = P \times 1.3225$
P = ₹10,000

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101. A train 240 m long crosses a 360 m platform in 30 seconds. Find its speed.

Answer: 20 m/s

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Explanation:
Total distance = 240 + 360 = 600 m
Speed = 600/30 = 20 m/s

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102. A can complete a work in 15 days, B in 20 days, and C in 30 days. How long together?

Answer: 6⅔ days

Explanation:
Combined rate = 1/15 + 1/20 + 1/30
LCM = 60
= 4/60 + 3/60 + 2/60 = 9/60 = 3/20
Time = 20/3 = 6⅔ days

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103. A and B invest ₹40,000 and ₹60,000 for 12 and 8 months respectively. Find their profit ratio.

Answer: 1:1

Explanation:
Profit ratio = Capital × Time
A = 40,000 × 12 = 480,000
B = 60,000 × 8 = 480,000
Ratio = 1:1

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104. A vessel contains 60 L of milk. 12 L is removed and replaced with water. This is repeated once. How much milk remains?

Answer: 38.4 L

Explanation:
After one replacement, milk remaining = 60 × 48/60 = 48 L.
After second replacement = 48 × 48/60 = 38.4 L

---

105. Two numbers are in the ratio 5:7 and their LCM is 420. Find their HCF.

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Answer: 12

Explanation:
Numbers = 5x and 7x.
LCM = 35x = 420
x = 12.
Therefore HCF = 12

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106. The average of 8 numbers is 24. If one number 32 is replaced by 48, find the new average.

Answer: 26

Explanation:
Original total = 8 × 24 = 192
New total = 192 − 32 + 48 = 208
New average = 208/8 = 26

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107. A trader gives 20% discount and still makes 25% profit. If CP is ₹800, find marked price.

Answer: ₹1,200

Explanation:
SP = 800 × 125/100 = ₹1,000
After 20% discount, SP = 80% of MP.
MP = 1000/0.8 = ₹1,250

Note: The correct answer is ₹1,250, not ₹1,200.

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108. A man travels half the distance at 40 km/h and half at 60 km/h. Find average speed.

Answer: 48 km/h

Explanation:
For equal distances:
Average speed = 2ab/(a+b)
= 2 × 40 × 60 / 100
= 48 km/h

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109. A boat's downstream speed is 18 km/h and upstream speed is 10 km/h. Find stream speed.

Answer: 4 km/h

Explanation:
Stream speed = (Downstream − Upstream)/2
= (18 − 10)/2 = 4 km/h

---

110. A sum is divided among A, B and C in ratio 2:3:5. If C gets ₹4,500, find total sum.

Answer: ₹9,000

Explanation:
5 parts = ₹4,500
1 part = ₹900
Total parts = 10
Total = 10 × 900 = ₹9,000

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111. A pipe fills a tank in 12 hours and another in 18 hours. A drain empties it in 36 hours. How long to fill together?

Answer: 9 hours

Explanation:
Net rate = 1/12 + 1/18 − 1/36
= 3/36 + 2/36 − 1/36
= 4/36 = 1/9
Therefore time = 9 hours

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112. If $x + 1/x = 5$, find $x^2 + 1/x^2$.

Answer: 23

Explanation:
$(x + 1/x)^2 = x^2 + 2 + 1/x^2$

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25 = x^2 + 2 + 1/x^2
Therefore = 25 − 2 = 23

---

113. If a + b = 12 and ab = 35, find $a^2 + b^2$.

Answer: 74

Explanation:
$a^2 + b^2 = (a+b)^2 - 2ab$
$= 12^2 - 2(35)$
= 144 − 70 = 74

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114. Solve: $2x^2 - 7x + 3 = 0$. Find the roots.

Answer: 3 and 1/2

Explanation:
$2x^2 - 7x + 3$
$= (2x - 1)(x - 3)$
Therefore x = 1/2, 3

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115. Find the next number: 2, 6, 12, 20, 30, __.

Answer: 42

Explanation:
Differences are 4, 6, 8, 10.
Next difference = 12.
30 + 12 = 42

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116. Find the next number: 5, 11, 23, 47, __.

Answer: 95

Explanation:
Pattern: ×2 + 1
47 × 2 + 1 = 95

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117. A number leaves remainder 5 when divided by 7. What remainder does its square leave when divided by 7?

Answer: 4

Explanation:
Number ≡ 5 (mod 7)
Square ≡ 5^2 = 25
25 ÷ 7 leaves remainder 4

---

118. Find the least number which leaves remainder 3 when divided by 5, 6 and 8.

Answer: 123

Explanation:
Number − 3 must be divisible by 5, 6 and 8.
LCM = 120
Number = 120 + 3 = 123

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119. Find the number of trailing zeros in 100!.

Answer: 24

Explanation:
Trailing zeros come from factors of 10 = 2 × 5.
Number of 5s = floor(100/5) + floor(100/25)
= 20 + 4 = 24

---

120. How many distinct arrangements can be made from the letters of LEVEL?

Answer: 30

Explanation:
LEVEL has 5 letters.
L appears twice and E appears twice.

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Arrangements = 5!/(2! × 2!)
= 120/4 = 30

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121. How many ways can 5 people be seated in a row?

Answer: 120

Explanation:
Number of arrangements = 5!
= 5 × 4 × 3 × 2 × 1 = 120

---

122. How many ways can 3 people be selected from 8 people?

Answer: 56

Explanation:
8C3 = 8!/(3!5!)
= (8 × 7 × 6)/(3 × 2 × 1)
= 56

---

123. A bag has 4 red, 5 blue and 3 green balls. Probability of drawing two blue balls without replacement?

Answer: 5/33

Explanation:
Total balls = 12.
Probability first blue = 5/12.
After one blue, 4 blue remain out of 11.
Probability = 5/12 × 4/11 = 5/33

---

124. Two dice are rolled. Probability that the sum is 9?

Answer: 1/9

Explanation:
Combinations giving 9:
(3,6), (4,5), (5,4), (6,3) → 4 outcomes.

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Total = 36.
Probability = 4/36 = 1/9

---

125. A committee of 4 is selected from 6 men and 5 women. How many committees have exactly 2 women?

Answer: 150

Explanation:
Choose 2 men from 6 and 2 women from 5.
6C2 × 5C2 = 15 × 10 = 150

---

126. A clock gains 5 minutes every hour. If set correctly at 8 AM, what time will it show at 8 PM?

Answer: 9:00 PM

Explanation:
12 hours × 5 minutes = 60 minutes gained.
So at actual 8 PM, clock shows 9 PM.

---

127. At what time between 4 and 5 o'clock are the hands opposite?

Answer: 4:54 6/11

Explanation:
At 4:00, the hour hand is 120° ahead.
For opposite position, relative movement required = 180° − 120° = 60°.
Minute hand gains 5.5° per minute.
60 ÷ 5.5 = 10 10/11 minutes.

Therefore: 4:10 10/11, not 4:54 6/11.

---

128. Find the angle between clock hands at 7:20.

Answer: 100°

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Explanation:
Hour hand position = 30×7 + 0.5×20 = 220°
Minute hand = 20×6 = 120°
Difference = 100°
Therefore = 100°

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129. A 20% increase followed by a 20% decrease results in what net change?

Answer: 4% decrease

Explanation:
Assume original = 100.
After 20% increase = 120.
After 20% decrease = 120 × 80/100 = 96.
Net change = 4% decrease

---

130. If A's salary is 25% more than B's, B's salary is what percent less than A's?

Answer: 20%

Explanation:
Assume B = 100.
A = 125.
Difference = 25.
Percentage less = 25/125 × 100 = 20%

---

131. A population increases 10% annually. If current population is 48,400, what was it 2 years ago?

Answer: 40,000

Explanation:
Current = Original × 1.1^2
48,400 = Original × 1.21
Original = 48,400/1.21 = 40,000

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132. A person buys shares at ₹120 and sells at ₹138 after paying 2% brokerage on sale value. Find approximate net profit percentage.

Answer: 12.5%

Explanation:
Brokerage = 2% of 138 = ₹2.76
Net SP = ₹135.24
Profit = 135.24 − 120 = ₹15.24
Profit% = 15.24/120 × 100 ≈ 12.7%, approximately 12.5%

---

133. A shopkeeper mixes two varieties costing ₹60/kg and ₹90/kg in ratio 2:3. Find mixture cost per kg.

Answer: ₹78/kg

Explanation:
Weighted average = (60×2 + 90×3)/(2+3)
= (120 + 270)/5
= ₹78/kg

---

134. How much water should be added to 20 L milk to make milk:water = 4:1?

Answer: 5 L

Explanation:
4 parts milk = 20 L
1 part = 5 L
Therefore water needed = 5 L

---

135. A 40 L solution has acid and water in ratio 3:2. How much water must be added to make ratio 3:4?

Answer: 16 L

Explanation:
Acid = 40 × 3/5 = 24 L
Water = 16 L

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For ratio 3:4, 24 L acid needs 32 L water.
Additional water = 32 − 16 = 16 L

---

136. A can do 1/3 of a job in 5 days. How many days for the whole job?

Answer: 15 days

Explanation:
1/3 work = 5 days
Complete work = 5 × 3 = 15 days

---

137. A and B together finish work in 10 days. A alone takes 15 days. How long does B alone take?

Answer: 30 days

Explanation:
Together = 1/10
A = 1/15
B = 1/10 − 1/15
= 3/30 − 2/30 = 1/30
Therefore B takes 30 days

---

138. A contractor hires 12 workers for 18 days. After 6 days, 4 workers leave. How many additional days are needed?

Answer: 18 days

Explanation:
Total work = 12 × 18 = 216 worker-days.
Work completed in 6 days = 12 × 6 = 72.
Remaining = 144 worker-days.
Workers remaining = 8.
Days = 144/8 = 18 days

---

139. A train traveling at 72 km/h crosses a man running at 18 km/h in the opposite direction. If train length is 270 m, time to cross him?

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Answer: 6 seconds

Explanation:
Relative speed = 72 + 18 = 90 km/h
90 × 5/18 = 25 m/s
Time = 270/25 = 10.8 seconds

Correct answer: 10.8 seconds.

---

140. Two trains 180 m and 220 m long travel in opposite directions at 54 and 72 km/h. Time to cross each other?

Answer: 12 seconds

Explanation:
Total distance = 180 + 220 = 400 m
Relative speed = 54 + 72 = 126 km/h
= 126 × 5/18 = 35 m/s
Time = 400/35 = 11.43 seconds

Correct answer: approximately 11.43 seconds.

---

141. A man rows 24 km downstream in 2 hours and returns in 3 hours. Find speed of boat in still water.

Answer: 10 km/h

Explanation:
Downstream speed = 24/2 = 12 km/h
Upstream speed = 24/3 = 8 km/h
Boat speed = (12 + 8)/2 = 10 km/h

---

142. If 3x + 2y = 16 and x − y = 2, find x and y.

Answer: x = 4, y = 2

Explanation:
x − y = 2 → x = y + 2
Substitute: 3(y+2)+2y = 16
5y + 6 = 16

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y = 2
x = 4

---

143. If 2x − 3y = 7 and x + y = 5, find x and y.

Answer: x = 22/5, y = 3/5

Explanation:
x + y = 5 → x = 5 − y
2(5−y) − 3y = 7
10 − 5y = 7
y = 3/5
x = 22/5

---

144. The roots of $x^2 - 9x + 20 = 0$ are what?

Answer: 4 and 5

Explanation:
$x^2 - 9x + 20$
$= (x-4)(x-5)$
Therefore x = 4 or 5

---

145. If $\log_2 x = 5$, find x.

Answer: 32

Explanation:
$\log_2 x = 5$ means $2^5 = x$.
Therefore x = 32

---

146. If 3^(x+1) = 81, find x.

Answer: 3

Explanation:
$81 = 3^4$

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Therefore x + 1 = 4
x = 3

---

147. Simplify: $(2^5 \times 2^3) / 2^4$.

Answer: 16

Explanation:
For same bases, add powers when multiplying and subtract when dividing.
= 2^(5+3−4)
= 2^4 = 16

---

148. If 1/x + 1/y = 1/6 and x = 12, find y.

Answer: 12

Explanation:
1/12 + 1/y = 1/6
1/y = 1/6 − 1/12 = 1/12
Therefore y = 12

---

149. A man spends 80% of his income. Income rises 25% and expenditure rises 20%. If initial income is ₹40,000, by what percentage do savings increase?

Answer: 45%

Explanation:
Initial income = ₹40,000
Initial expenditure = 80% = ₹32,000
Initial savings = ₹8,000

New income = 40,000 × 1.25 = ₹50,000
New expenditure = 32,000 × 1.20 = ₹38,400
New savings = ₹11,600

Increase = ₹3,600
Percentage increase = 3,600/8,000 × 100 = 45%

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150. A product is marked 50% above CP and sold after successive discounts of 10% and 20%. Find profit percentage.

Answer: 8%

Explanation:
Assume CP = ₹100.
Marked Price = ₹150.
After 10% discount = ₹135.
After 20% discount = ₹108.
Profit = ₹8.
Profit% = 8%

---

151. A dealer buys an article for ₹2,400, spends ₹300 on repairs and sells it for ₹3,240. Find profit percentage.

Answer: 20%

Explanation:
Total CP = 2400 + 300 = ₹2,700
Profit = 3240 − 2700 = ₹540
Profit% = 540/2700 × 100 = 20%

---

152. A sum doubles in 8 years at simple interest. In how many years will it become 3 times?

Answer: 16 years

Explanation:
Doubling means interest = 100% of principal in 8 years.
For 3 times, interest = 200%.
Therefore time = 8 × 2 = 16 years

---

153. At simple interest, a sum becomes ₹1,440 in 3 years and ₹1,680 in 5 years. Find principal.

Answer: ₹1,080

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Explanation:
Difference in amount for 2 years = 1680 − 1440 = ₹240.
Annual interest = ₹120.
3-year interest = ₹360.
Principal = 1440 − 360 = ₹1,080

---

154. The compound interest on a sum for 2 years at 10% is ₹525. Find the principal.

Answer: ₹2,500

Explanation:
CI for 2 years at 10% = $P[(1.1)^2 - 1]$
= P × 0.21
0.21P = 525
P = ₹2,500

---

155. A population decreases 10% one year and increases 10% next year. Find net percentage change.

Answer: 1% decrease

Explanation:
Assume population = 100.
After 10% decrease = 90.
After 10% increase = 99.
Net = 1 less than original.
Therefore 1% decrease.

---

156. If A is 40% of B and B is 25% of C, A is what percent of C?

Answer: 10%

Explanation:
A = 40% of B = 0.4B
B = 25% of C = 0.25C
A = 0.4 × 0.25C = 0.10C
Therefore = 10%

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157. A number is first multiplied by 1.2 and then by 0.8. Find net percentage change.

Answer: 4% decrease

Explanation:
Overall multiplier = 1.2 × 0.8 = 0.96
Final value = 96% of original.
Decrease = 4%

---

158. A and B have efficiencies in ratio 3:2. If A takes 12 days, how long does B take?

Answer: 18 days

Explanation:
Time is inversely proportional to efficiency.
A:B efficiency = 3:2
B's time = 12 × 3/2 = 18 days

---

159. A, B and C can complete work in 12, 15 and 20 days respectively. How long together?

Answer: 5 days

Explanation:
Combined rate = 1/12 + 1/15 + 1/20
LCM = 60
= 5/60 + 4/60 + 3/60 = 12/60 = 1/5
Time = 5 days

---

160. A leak empties a full tank in 20 hours while a pipe fills it in 12 hours. If both are open, how long to fill?

Answer: 30 hours

Explanation:
Net rate = 1/12 − 1/20

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= 5/60 − 3/60 = 2/60 = 1/30
Time = 30 hours

---

161. A pipe fills 1/4 tank in 3 hours. Another fills 1/3 tank in 4 hours. How long together for full tank?

Answer: 6 hours

Explanation:
First pipe rate = 1/12 tank/hour.
Second pipe rate = 1/12 tank/hour.
Together = 1/6 tank/hour.
Time = 6 hours

---

162. A man covers 60 km at 30 km/h and next 60 km at 60 km/h. Find average speed.

Answer: 40 km/h

Explanation:
Total distance = 120 km.
Time = 60/30 + 60/60 = 2 + 1 = 3 hours.
Average speed = 120/3 = 40 km/h

---

163. A cyclist increases speed from 20 to 25 km/h and saves 12 minutes on a journey. Find distance.

Answer: 20 km

Explanation:
Time saved = 12 minutes = 0.2 hour.
D/20 − D/25 = 0.2
D(1/20 − 1/25) = 0.2
D/100 = 0.2
D = 20 km

---

164. Two numbers have sum 45 and product 494. Find the numbers.

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Answer: 19 and 26

Explanation:
We need two numbers whose sum is 45 and product is 494.
19 + 26 = 45
19 × 26 = 494
Therefore = 19 and 26

---

165. If $x - 1/x = 3$, find $x^2 + 1/x^2$.

Answer: 11

Explanation:
$(x - 1/x)^2 = x^2 - 2 + 1/x^2$
$9 = x^2 - 2 + 1/x^2$
Therefore $x^2 + 1/x^2 = 11$

---

166. If a:b = 3:4 and b:c = 8:9, find a:b:c.

Answer: 6:8:9

Explanation:
First ratio = 3:4.
Make b = 8 by multiplying by 2 → 6:8.
Second ratio = 8:9.
Therefore = 6:8:9

---

167. Find the greatest common divisor of 84, 126 and 210.

Answer: 42

Explanation:
84 = 42 × 2
126 = 42 × 3
210 = 42 × 5
Therefore GCD = 42

---

168. Find the smallest number that is divisible by 12, 15 and 18.

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Answer: 180

Explanation:
Prime factors:
$12 = 2^2 \times 3$
$15 = 3 \times 5$
$18 = 2 \times 3^2$
LCM = $2^2 \times 3^2 \times 5 = 180$

---

169. How many integers from 1 to 500 are divisible by 3 but not by 5?

Answer: 133

Explanation:
Multiples of 3 = floor(500/3) = 166.
Multiples of both 3 and 5 = multiples of 15 = floor(500/15) = 33.
Required = 166 − 33 = 133

---

170. How many positive divisors does 360 have?

Answer: 24

Explanation:
$360 = 2^3 \times 3^2 \times 5^1$
Number of divisors = (3+1)(2+1)(1+1)
= 4 × 3 × 2 = 24

---

171. Find the sum of all divisors of 28.

Answer: 56

Explanation:
Divisors = 1, 2, 4, 7, 14, 28
Sum = 1 + 2 + 4 + 7 + 14 + 28 = 56

---

172. If n is a positive integer and n(n+1) is divisible by 2, why?

Answer: One of two consecutive integers is always even.

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Explanation:
Any two consecutive integers contain one even number.
Every even number is divisible by 2.
Therefore n(n+1) is always divisible by 2.

---

173. How many 4-digit numbers can be formed using 1,2,3,4,5 without repetition?

Answer: 120

Explanation:
5 choices for first digit, 4 for second, 3 for third and 2 for fourth.
5 × 4 × 3 × 2 = 120

---

174. How many 4-digit even numbers can be formed from 1,2,3,4,5 without repetition?

Answer: 48

Explanation:
Last digit must be even: 2 choices (2 or 4).
Remaining three positions: 4 × 3 × 2 = 24.
Total = 2 × 24 = 48

---

175. How many diagonals does a polygon with 12 sides have?

Answer: 54

Explanation:
Formula = n(n−3)/2
= 12 × 9/2
= 54

---

176. The interior angle of a regular polygon is 150°. How many sides does it have?

Answer: 12

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Explanation:
Interior angle = (n−2)180/n
150n = 180n − 360
30n = 360
n = 12

---

177. A rectangle has diagonal 13 cm and length 12 cm. Find breadth.

Answer: 5 cm

Explanation:
Using Pythagoras:
$13^2 = 12^2 + b^2$
169 = 144 + b^2
$b^2 = 25$
b = 5 cm

---

178. A right triangle has legs 9 cm and 12 cm. Find hypotenuse.

Answer: 15 cm

Explanation:
Hypotenuse² = $9^2 + 12^2$
= 81 + 144 = 225
Hypotenuse = 15 cm

---

179. A cylinder has radius 7 cm and height 10 cm. Using π = 22/7, find volume.

Answer: 1,540 cm^3

Explanation:
Volume = $\pi r^2 h$
= 22/7 × 7^2 × 10
= 22 × 7 × 10
= 1,540 cm^3

---

180. A sphere has radius 3 cm. Using π = 22/7, find its volume.

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Answer: 792/7 cm^3 ≈ 113.14 cm^3

Explanation:
Volume = $4/3\pi r^3$
= $4/3 \times 22/7 \times 27$
= 792/7
≈ 113.14 cm^3

---

181. If the radius of a circle increases by 10%, by what percentage does its area increase?

Answer: 21%

Explanation:
Area depends on r^2.
New radius = 1.1r
New area = $(1.1)^2$ = 1.21 times original.
Increase = 21%

---

182. If the side of a cube increases by 20%, by what percentage does volume increase?

Answer: 72.8%

Explanation:
Volume depends on $side^3$.
New side = 1.2s
New volume = 1.2^3 = 1.728
Increase = 72.8%
Therefore = 72.8%

---

183. A class average rises from 68 to 70 after a student scoring 50 is replaced. If class has 20 students, what is the new student's score?

Answer: 90

Explanation:
Original total = 68 × 20 = 1,360
New total = 70 × 20 = 1,400

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Increase in total = 40
New score = 50 + 40 = 90

---

184. Average age of 5 people is 24. A sixth joins and average becomes 25. Find sixth person's age.

Answer: 30 years

Explanation:
Original total = 5 × 24 = 120
New total = 6 × 25 = 150
Sixth person's age = 150 − 120 = 30 years

---

185. The average of 7 consecutive integers is 31. Find the largest integer.

Answer: 34

Explanation:
For consecutive numbers, the average is the middle number.
Seven numbers are: 28, 29, 30, 31, 32, 33, 34.
Largest = 34

---

186. A batsman has average 48 after 10 innings. What score is needed in 11th to raise average to 50?

Answer: 70

Explanation:
Current total = 48 × 10 = 480
Required total = 50 × 11 = 550
Required score = 550 − 480 = 70

---

187. A company has 60% male employees. If there are 240 female employees, find total employees.

Answer: 600

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Explanation:
Female = 40% of total.
40% = 240
Total = 240 × 100/40 = 600

---

188. In a survey, 70% like tea, 60% like coffee, and 40% like both. What percentage like at least one?

Answer: 90%

Explanation:
Using: $A \cup B = A + B - A \cap B$
$= 70 + 60 - 40 = 90\%$

---

189. In a group of 100 people, 55 know Python, 45 know Java, and 20 know both. How many know neither?

Answer: 20

Explanation:
At least one = 55 + 45 − 20 = 80
Neither = 100 − 80 = 20

---

190. A probability is 0.35. What is probability of the complementary event?

Answer: 0.65

Explanation:
Complementary probability = 1 − P(Event)
= 1 − 0.35 = 0.65

---

191. Three coins are tossed. Probability of exactly two heads?

Answer: 3/8

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Explanation:
Possible outcomes = $2^3 = 8$.
Exactly two heads: HHT, HTH, THH → 3 outcomes.
Probability = 3/8

---

192. A card is drawn from a standard deck. Probability of a king or queen?

Answer: 2/13

Explanation:
Kings = 4
Queens = 4
Favourable cards = 8
Total cards = 52
Probability = 8/52 = 2/13

---

193. A box contains 6 defective and 14 good items. Two are selected. Probability both are good?

Answer: 91/190

Explanation:
Total = 20.
Probability first good = 14/20.
Second good = 13/19.
Probability = 14/20 × 13/19
= 182/380 = 91/190

---

194. A password has 3 distinct digits from 0–9 without repetition. How many passwords?

Answer: 720

Explanation:
First digit: 10 choices
Second: 9 choices
Third: 8 choices
Total = 10 × 9 × 8 = 720

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---

195. A number is chosen from 1 to 100. Probability it is a multiple of 7?

Answer: 7/50

Explanation:
Multiples of 7 from 1 to 100 = 14.
Probability = 14/100 = 7/50

---

196. If today is Monday, what day will it be 100 days later?

Answer: Wednesday

Explanation:
100 ÷ 7 gives remainder 2.
Two days after Monday = Wednesday

---

197. A clock is 10 minutes slow at 6 PM and loses 2 minutes every hour. What will it show at 10 PM?

Answer: 9:42 PM

Explanation:
From 6 PM to 10 PM = 4 hours.
Additional loss = 4 × 2 = 8 minutes.
Total loss = 10 + 8 = 18 minutes.
At 10 PM, clock shows 9:42 PM.

---

198. A person invests ₹20,000 at 8% simple interest and another ₹30,000 at 10% simple interest. Find total interest earned in 2 years.

Answer: ₹7,600

Explanation:
First investment:
SI = 20,000 × 8 × 2/100 = ₹3,200

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Second investment:
SI = 30,000 × 10 × 2/100 = ₹6,000

Total SI = 3,200 + 6,000 = ₹9,200

Correct answer: ₹9,200.

---

199. A work is completed by 10 men in 24 days. After 8 days, 2 men leave. How many more days are required?

Answer: 20 days

Explanation:
Total work = 10 × 24 = 240 man-days.
Work completed in 8 days = 10 × 8 = 80 man-days.
Remaining work = 240 − 80 = 160 man-days.
Remaining workers = 8.
Required days = 160 ÷ 8 = 20 days

---

200. If the average of 12 numbers is 25 and the average of the first 7 is 22, find the average of the remaining 5.

Answer: 29.2

Explanation:
Total of 12 numbers = 12 × 25 = 300.
Total of first 7 = 7 × 22 = 154.
Total of remaining 5 = 300 − 154 = 146.
Average = 146 ÷ 5 = 29.2

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QUICK FORMULA SHEET

Percentage:
Percentage = (Part / Whole) × 100

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Profit:
Profit = SP − CP
Profit% = (Profit / CP) × 100

Loss:
Loss = CP − SP
Loss% = (Loss / CP) × 100

Simple Interest:
SI = P × R × T / 100

Compound Interest:
$A = P(1 + R/100)^n$

Average:
Average = Sum / Number of values

Speed:
Speed = Distance / Time

Distance:
Distance = Speed × Time

Time:
Time = Distance / Speed

Work:
Work = Efficiency × Time

Probability:
Probability = Favourable outcomes / Total outcomes

Area of Rectangle:
Length × Breadth

Area of Triangle:
½ × Base × Height

Area of Circle:
πr^2

Circumference:
$2\pi r$

Volume of Cube:
$Side^3$

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Volume of Cylinder:
$\pi r^2 h$

Pythagoras Theorem:
$a^2 + b^2 = c^2$

Number of diagonals in polygon:
$n(n-3)/2$

Permutations:
$nPr = n!/(n-r)!$

Combinations:
$nCr = n!/[r!(n-r)!]$

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